

# Forbidden Induced Subgraphs for Bounded Shrub-Depth and the Expressive Power of MSO

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## Abstract

The graph parameter shrub-depth is a dense analog of tree-depth. We characterize classes of bounded shrub-depth by forbidden induced subgraphs. The obstructions are well-controlled flips of large half-graphs and of disjoint unions of many long paths. Applying this characterization, we show that on every hereditary class of unbounded shrub-depth, MSO is more expressive than FO. This confirms a conjecture of [Gajarský and Hliněný; LMCS 2015] who proved that on classes of bounded shrub-depth FO and MSO have the same expressive power. Combined, the two results fully characterize the hereditary classes on which FO and MSO coincide, answering an open question by [Elberfeld, Grohe, and Tantau; LICS 2012].

Our work is inspired by the notion of stability from model theory. A graph class  $\mathcal{C}$  is MSO-stable, if no MSO-formula can define arbitrarily long linear orders in graphs from  $\mathcal{C}$ . We show that a hereditary graph class is MSO-stable if and only if it has bounded shrub-depth. As a key ingredient, we prove that every hereditary class of unbounded shrub-depth FO-interprets the class of all paths. This improves upon a result of [Ossona de Mendez, Pilipczuk, and Siebertz; Eur. J. Comb. 2025] who showed the same statement for FO-transductions instead of FO-interpretations.

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## 1 Introduction

The main result of this paper is the following Theorem 1 yielding various characterizations for hereditary graph classes of bounded shrub-depth, in terms of forbidden induced subgraphs, (counting) monadic second-order logic (MSO and CMSO), and first-order logic (FO). All appearing notions will be motivated and defined in the remainder of this introduction.

► **Theorem 1.** *For every hereditary graph class  $\mathcal{C}$ , the following are equivalent.*

1.  $\mathcal{C}$  has bounded shrub-depth.
2. There is a  $t \in \mathbb{N}$  such that  $\mathcal{C}$  excludes all flipped half-graphs of order  $t$  and all flipped  $tP_t$ .
3. There is a  $t \in \mathbb{N}$  such that  $\mathcal{C}$  excludes all flipped half-graphs of order  $t$  and all flipped  $3P_t$ .
4.  $\mathcal{C}$  is MSO-stable.
5.  $\mathcal{C}$  is monadically MSO-stable.
6.  $\mathcal{C}$  is CMSO-stable.
7.  $\mathcal{C}$  is monadically CMSO-stable.
8.  $\mathcal{C}$  does not 1-dimensionally FO-interpret the class of all paths.
9. FO and MSO have the same expressive power on  $\mathcal{C}$ .



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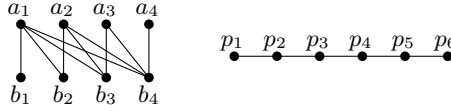


## Stability

We start motivating Theorem 1 by introducing the model-theoretic notion of *stability*, a context in which the graph parameter shrub-depth will naturally arise. Originating in the 70s and pioneered by Shelah, stability theory is a prolific branch of model theory which seeks to classify the complexity of theories (or in our case: graph classes). There, stability is the most important dividing line, which separates the well-behaved *stable* classes from the complex *unstable* ones. Intuitively, a graph class  $\mathcal{C}$  is stable, if one cannot define arbitrarily large linear orders in  $\mathcal{C}$  using logical formulas. More precisely, for a logic  $\mathcal{L} \in \{\text{FO}, \text{MSO}, \text{CMSO}\}$ , an  $\mathcal{L}$ -formula  $\varphi(\bar{x}, \bar{y})$ , and a graph class  $\mathcal{C}$ , we say  $\varphi$  has the *order-property* on  $\mathcal{C}$ , if for every  $\ell \in \mathbb{N}$  there is a graph  $G \in \mathcal{C}$  and a sequence  $\bar{a}_i, \dots, \bar{a}_\ell$  of tuples of vertices of  $G$ , such that for all  $i, j \in [\ell]$

$$G \models \varphi(\bar{a}_i, \bar{a}_j) \iff i \leq j.$$

For example the FO-formula  $\varphi(x, y)$  expressing “the neighborhood of  $x$  is a superset of the neighborhood of  $y$ ” has the order-property on the class of all half-graphs, as it orders the sequence  $a_1, \dots, a_t$  in the half-graph  $H_t$  of order  $t$  for every  $t \in \mathbb{N}$ , as depicted in Figure 1.



■ **Figure 1** On the left: the half-graph of order 4 (denoted as  $H_4$ ) with vertices  $\{a_1, \dots, a_4, b_1, \dots, b_4\}$  and edges between  $a_i$  and  $b_j$  if  $i \leq j$ . On the right: the 6-vertex path (denoted as  $P_6$ ).

Similarly, the MSO-formula  $\psi(x_1x_2, y_1y_2)$  expressing “ $x_1$  and  $x_2$  are not connected after deleting  $y_1$ ” has the order-property on the class of all paths, as it orders the sequence of 2-tuples  $p_1p_t, p_2p_t, \dots, p_tp_t$  in the  $t$ -vertex path  $P_t$  for every  $t \in \mathbb{N}$ . A graph class  $\mathcal{C}$  is  $\mathcal{L}$ -stable if no  $\mathcal{L}$ -formula has the order-property on  $\mathcal{C}$ , and  $\mathcal{L}$ -unstable otherwise. The class of all half-graphs (all paths) is the arguably simplest example of an FO-unstable (MSO-unstable) graph class.

Apart from its extensive study on infinite structures in model theory, FO-stability has recently gained a lot of attention in finite model theory, in particular in structural and algorithmic graph theory. Podewski and Ziegler [29] and Adler and Adler [1] observed that on *monotone*<sup>1</sup> graph classes, FO-stability coincides with the combinatorial property of being *nowhere dense*. Nowhere dense classes are very general classes of sparse graphs [26], enjoying strong combinatorial and algorithmic properties, such as fixed-parameter tractable (fpt) FO model checking [18]. The equivalence of FO-stability and nowhere denseness in monotone classes elegantly bridges the fields of model theory and structural graph theory. It has prompted the question whether also *hereditary*<sup>2</sup> FO-stable graph classes are combinatorially well-behaved. In the hereditary setting, FO-stability significantly generalizes nowhere denseness: for example the class of all cliques and the class of all map graphs are both FO-stable but not nowhere dense. A current research program has uncovered multiple natural combinatorial characterizations of hereditary FO-stable classes [8, 10, 15, 5] and shown that they also admit fpt FO model checking [15, 8, 9].

<sup>1</sup> A graph class is *monotone*, if it is closed under taking subgraphs.

<sup>2</sup> A graph class is *hereditary*, if it is closed under taking induced subgraphs.

In contrast to the rich literature on FO-stability, we are not aware of any previous works studying its natural restriction MSO-stability. We attribute this to the fact that the compactness theorem, which is the main tool for working on *infinite* structures where stability originates, fails for MSO. Now, the recent successes in the study of FO-stable classes of *finite* graphs raise the question whether also MSO-stability can be understood through the lens of structural graph theory. In this work we show that this is indeed the case, by proving the following.

► **Theorem 2.** *A hereditary graph class is MSO-stable if and only if it has bounded shrub-depth.*

The folklore fact that every monotone class of bounded shrub-depth also has bounded tree-depth yields the following corollary.

► **Corollary 3.** *A monotone graph class is MSO-stable if and only if it has bounded tree-depth.*

## Shrub-Depth

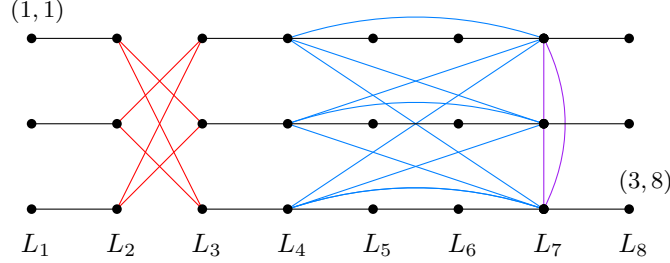
*Shrub-depth* is a parameter for graph classes introduced in [16]. It generalizes tree-depth to dense classes and can be seen as a bounded depth analog of clique- and rank-width, similar to how tree-depth is a bounded depth analog of tree-width. Classes of bounded shrub-depth are extremely well-behaved from algorithmic, combinatorial, and logical points of view. For instance, they admit quadratic time isomorphism testing [27] and fpt MSO model checking with an elementary dependence on the size of the input formula [14, 16]; they can be characterized by vertex minors and FO-transductions [28]; and FO and MSO have the same expressive power on these classes [14].

The above examples show that the structure side of bounded shrub-depth is well charted. Indeed, we prove the direction “bounded shrub-depth implies MSO-stability” of Theorem 2 using mainly existing tools. In contrast, proving the “hereditary and unbounded shrub-depth implies MSO-instability” direction requires insights about the non-structure side of shrub-depth (i.e., the properties shared by all classes that have unbounded shrub-depth). Up until now, the picture here was much more vague: In [17] it is proved that for every  $d \in \mathbb{N}$  there exists a finite set of graphs  $\mathcal{F}_d$  such that a graph has shrub-depth at most  $d$  if and only if it excludes all of  $\mathcal{F}_d$  as induced subgraphs. This result is non-constructive and does not reveal the concrete sets  $\mathcal{F}_d$ . In [21] and [28] it is shown that the class of all paths can be produced from any class  $\mathcal{C}$  of unbounded shrub-depth by taking vertex-minors and also by an FO-transduction, respectively. The high expressive power and irreversibility of vertex-minors and transductions limits our ability to draw conclusions about the structure of the original class  $\mathcal{C}$  from these two results. In this paper, we improve the non-structure situation by providing a characterization of classes of bounded shrub-depth by explicitly listing forbidden induced subgraphs.

## Forbidden Induced Subgraphs

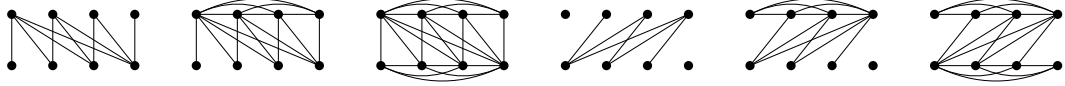
We will briefly introduce the notions needed to state the obstructions. For two graphs  $G$  and  $H$  on the same vertex set and a partition  $\mathcal{P}$  of  $V(G)$ , we say  $H$  is a  $\mathcal{P}$ -flip of  $G$  if it can be obtained by complementing the edge relation between pairs of parts of  $\mathcal{P}$  (see Figure 2). We refer to the preliminaries for formal definitions. Our obstructions for bounded shrub-depth will be well-controlled flips of disjoint unions of long paths and of large half-graphs, as made precise in the following two definitions and their accompanying Figures 2 and 3.

► **Definition 4.**  $P_t$  is the  $t$ -vertex path and  $mP_t$  is the disjoint union of  $m$  many  $P_t$ , with vertices  $[m] \times [t]$ , where  $(i, j)$  is the  $j$ th vertex on the  $i$ th path. A flipped  $mP_t$  is an  $\mathcal{L}$ -flip of  $mP_t$  for the partition  $\mathcal{L} = \{L_1, \dots, L_t\}$  of the paths into layers, where  $L_j = \{(1, j), \dots, (m, j)\}$  contains the  $j$ th vertices of all the paths for  $j \in [t]$ .



■ **Figure 2** A flipped  $3P_8$ . More precisely the depicted graph is an  $\mathcal{L}$ -flip of  $3P_8$  with  $\mathcal{L} = \{L_1, \dots, L_8\}$ , where the following parts were flipped:  $L_2$  with  $L_3$  (red),  $L_4$  with  $L_7$  (blue),  $L_7$  with  $L_8$  (purple).

► **Definition 5.** The half-graph of order  $t$  (denoted as  $H_t$ ) is the bipartite graph on vertices  $A = \{a_1, \dots, a_t\}$  and  $B = \{b_1, \dots, b_t\}$  where  $a_i$  and  $b_j$  are adjacent if and only if  $i \leq j$ . A flipped  $H_t$  is a  $\mathcal{P}$ -flip of  $H_t$ , for the partition  $\mathcal{P} = \{A, B\}$ .



■ **Figure 3** All flipped  $H_4$ s (up to isomorphism). Replicated with permission from [24, Fig. 2.5].

We are now ready to state our characterization.

► **Theorem 6.** A graph class  $\mathcal{C}$  has bounded shrub-depth if and only if there is  $t \in \mathbb{N}$  such that  $\mathcal{C}$  excludes all flipped  $H_t$  and all flipped  $tP_t$  as induced subgraphs.

Moreover, our proofs show that in Theorem 6, one can replace  $tP_t$  with  $3P_t$ . While the stated  $tP_t$  variant of the theorem is more useful for hardness proofs, the  $3P_t$  variant is a step towards finding the simplest obstructions that cause unbounded shrub-depth. It remains open whether the theorem also holds for  $2P_t$ , but we know it fails for  $1P_t$ , as every graph on  $t$  vertices is a flipped  $1P_t$ .

## Interpretations and Monadic Stability

In order to prove the stability characterization (Theorem 2) from the induced subgraph characterization (Theorem 6), we visit another model-theoretic concept called *interpretations*. For a logic  $\mathcal{L}$  a 1-dimensional<sup>3</sup>  $\mathcal{L}$ -interpretation  $I$  is defined by two  $\mathcal{L}$ -formulas: a domain formula  $\delta(x)$  and an irreflexive, symmetric edge formula  $\varphi(x, y)$ . It maps each graph  $G$  to the graph  $H := I(G)$  with vertex set  $V(H) := \{v \in V(G) : G \models \delta(v)\}$  and edges  $E(H) := \{(u, v) \in V(H)^2 : G \models \varphi(u, v)\}$ . We say a graph class  $\mathcal{C}$  *interprets* a graph class  $\mathcal{D}$ , if there is an interpretation  $I$  such that  $\mathcal{D} \subseteq \{I(G) : G \in \mathcal{C}\}$ . Building on our forbidden induced subgraph characterization, we show the following.

<sup>3</sup> We will later also define and work with higher dimensional interpretations, whose formulas have tuples as free variables.

► **Theorem 7.** *A hereditary graph class has unbounded shrub-depth if and only if it 1-dimensionally FO-interprets the class of all paths.*

As we have already seen, the class of all paths is MSO-unstable. Theorem 7 can be used to lift instability to all hereditary classes of unbounded shrub-depth. This is a strengthening of a recent result by Ossona de Mendez, Pilipczuk, and Siebertz [28], which states that every class of unbounded shrub-depth *FO-transduces*<sup>4</sup> the class of all paths. Their result implies that every class of unbounded shrub-depth is *monadically MSO-unstable*: a graph class  $\mathcal{C}$  is monadically  $\mathcal{L}$ -stable if every coloring of  $\mathcal{C}$  is  $\mathcal{L}$ -stable. In general, monadic MSO-stability is more restrictive than MSO-stability. For instance the class  $\mathcal{K}$  of all 1-subdivided cliques is MSO-stable, but monadically MSO-unstable. When considering only hereditary classes, this example fails: the hereditary closure of  $\mathcal{K}$  contains every 1-subdivided graph and is FO-unstable (so in particular also monadically FO-unstable, MSO-unstable, and monadically MSO-unstable). Braunfeld and Laskowski showed that FO-stability and monadic FO-stability coincide in all hereditary classes [4]. We show that the same collapse happens for MSO and CMSO.

► **Theorem 8.** *For hereditary graph classes, the notions MSO-stability, monadic MSO-stability, CMSO-stability, and monadic CMSO-stability are all equivalent.*

## The Expressive Power of MSO

For a graph class  $\mathcal{C}$ , we say FO and MSO *have the same expressive power on  $\mathcal{C}$*  if for every MSO-sentence  $\varphi$ , there exists an FO-sentence  $\psi$  such that for every graph  $G \in \mathcal{C}$  we have  $G \models \varphi \Leftrightarrow G \models \psi$ . Otherwise, we say MSO is *more expressive* than FO on  $\mathcal{C}$ .

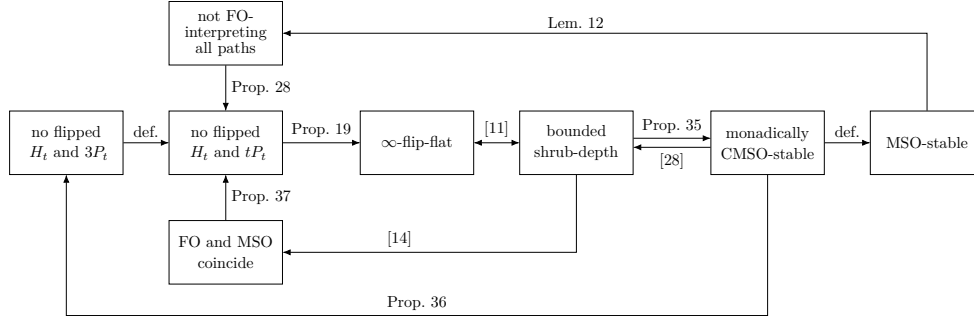
It was shown by Elberfeld, Grohe, and Tantau that for all monotone classes  $\mathcal{C}$ , FO and MSO have the same expressive power if and only if  $\mathcal{C}$  has bounded tree-depth [12]. As an open question, they asked for a characterization of the hereditary classes where FO and MSO coincide. As a dense analog of bounded tree-depth, bounded shrub-depth is a natural candidate here. Indeed, Gajarský and Hliněný showed that FO and MSO have the same expressive power on every class of bounded shrub-depth [14, Thm. 5.14]. However, they could not prove the reverse direction, which they attributed to a lack of known obstructions for classes of bounded shrub-depth. As an application of our forbidden induces subgraph characterization, we provide the missing part of the puzzle by showing that MSO is more expressive than FO on every hereditary class of unbounded shrub-depth. Together with the result by Gajarský and Hliněný, this completely characterizes the hereditary classes on which FO and MSO coincide.

► **Theorem 9.** *For every hereditary graph class  $\mathcal{C}$ , FO and MSO have the same expressive power on  $\mathcal{C}$  if and only if  $\mathcal{C}$  has bounded shrub-depth.*

## Overview of the Paper

We present an extended abstract of the paper. Proofs of statements marked with ♣ can be found in the full version [25] of the paper. Figure 4 shows the implications which comprise Theorem 1. The only notion in the figure that has not been discussed so far is

<sup>4</sup> A transduction first colors the input graph non-deterministically, and then applies a fixed 1-dimensional interpretation that can access these colors. For a single input graph  $G$ , it produces multiple (possibly isomorphic) output graphs: one for each coloring of  $G$ . Due to the access to an arbitrary coloring, transductions are more expressive than interpretations.



■ **Figure 4** A map of Theorem 1: characterizations of hereditary classes of bounded shrub-depth.

$\infty$ -*flip-flatness*. This is another characterization of shrub-depth with strong ties to stability theory, recently proved by Dreier, Mählmann, and Toruńczyk [11], which we define in the upcoming preliminaries (Section 2). The proofs of the individual implications are presented in Sections 3–6. We conclude with an outlook in Section 7, where we discuss potential strengthenings of our induced subgraph characterization and a second major dividing line from model theory, named *dependence*.

## 2 Preliminaries

We use  $[n]$  to refer to the set  $\{1, \dots, n\}$ . For an  $s$ -tuple  $\bar{a} = a_1 \dots a_s$ , we use  $|\bar{a}|$  to refer to its length  $s$  and  $\bar{a}[i]$  to refer to its  $i$ th component  $a_i$ .

**Colored Structures and Graphs.** A (*relational*) *signature*  $\Sigma$  is a set of relation symbols, each with an implicitly assigned arity. A structure  $G$  over  $\Sigma$  is called a  $\Sigma$ -*structure*. We denote by  $\Sigma^{(k)} := \Sigma \cup \{C_1, \dots, C_k\}$  the *expansion* of  $\Sigma$  by  $k$  many new unary predicates, which we can think of as colors. A single element is allowed to have multiple colors, but this will not be important. A  $\Sigma^{(k)}$ -structure  $G^+$  is a  $k$ -*coloring* of  $G$ , if  $G$  is the  $\Sigma$ -structure obtained by forgetting the color predicates in  $G^+$ . For a class of  $\Sigma$ -structures  $\mathcal{C}$ , we denote by  $\mathcal{C}^{(k)}$  the class of  $\Sigma^{(k)}$ -structures consisting of all the  $k$ -colorings of structures from  $\mathcal{C}$ .

A *graph* is a  $\Gamma$ -structure for the signature  $\Gamma := \{E\}$  consisting of a single binary relation (called *edge relation*) that is interpreted symmetrically and irreflexive. For any structure  $G$ , we use  $V(G)$  to refer to its universe: if  $G$  is a graph, then  $V(G)$  is its vertex set.

**Logic and Stability.** We assume familiarity with first-order logic (FO) and monadic second-order logic (MSO). As we model graphs as structures with a binary edge relation, MSO on graphs allows for quantification of vertex sets, but not of edge sets. This is also known as  $\text{MSO}_1$  in the literature. *Counting* monadic second-order logic (CMSO) extends MSO for all set variables  $X$  and  $0 \leq r < k$  by the cardinality constraint  $\text{card}_{r,k}(X)$  that holds true if and only if  $|X| \equiv r \pmod{k}$ . For a logic  $\mathcal{L} \in \{\text{FO}, \text{MSO}, \text{CMSO}\}$  and a signature  $\Sigma$ . We use  $\mathcal{L}[\Sigma]$  to refer to the set of  $\mathcal{L}$ -formulas over  $\Sigma$ . An  $\mathcal{L}[\Sigma]$ -formula  $\varphi(\bar{x}, \bar{y})$  has the  $\ell$ -*order-property* on a  $\Sigma$ -structure  $G$ , if there exists a sequence  $(\bar{a}_i)_{i \in [\ell]}$  of tuples of elements of  $G$ , such that for all  $i, j \in [\ell]$  we have  $G \models \varphi(\bar{a}_i, \bar{a}_j) \Leftrightarrow i \leq j$ . The formula  $\varphi$  has the *order-property* on a class of  $\Sigma$ -structures  $\mathcal{C}$ , if for every  $\ell \in \mathbb{N}$  there is  $G \in \mathcal{C}$  such that  $\varphi$  has the  $\ell$ -order-property on  $G$ . We call a class of  $\Sigma$ -structures  $\mathcal{L}$ -*stable*, if no  $\mathcal{L}[\Sigma]$ -formula has the order-property on  $\mathcal{C}$ . Moreover,  $\mathcal{C}$  is *monadically  $\mathcal{L}$ -stable* if for every  $k \in \mathbb{N}$ , the class  $\mathcal{C}^{(k)}$  of  $k$ -colorings from  $\mathcal{C}$  is  $\mathcal{L}$ -stable.

**Interpretations.** For a logic  $\mathcal{L}$ ,  $d \in \mathbb{N}$ , and signatures  $\Sigma_1$  and  $\Sigma_2$ , a  $d$ -dimensional  $\mathcal{L}$ -interpretation  $I$  from  $\Sigma_1$  to  $\Sigma_2$  consists of

- an  $\mathcal{L}[\Sigma_1]$ -formula  $\delta_I(\bar{x})$  called the *domain formula*,
- an  $\mathcal{L}[\Sigma_1]$ -formula  $\varphi_{I,R}(\bar{x}_1, \dots, \bar{x}_k)$  for every  $k$ -ary relation  $R$  in  $\Sigma_2$ ,

where  $\bar{x}$  and all  $\bar{x}_i$  are  $d$ -tuples. Given a  $\Sigma_1$ -structure  $G$ , we define the  $H := I(G)$  to be the  $\Sigma_2$ -structure with the universe  $V(H) := \{\bar{a} \in V(G)^d : G \models \delta_I(\bar{a})\}$  and relations  $R(H) := \{\bar{a}_1 \dots \bar{a}_k \in V(H)^k : G \models \varphi_{I,R}(\bar{a}_1, \dots, \bar{a}_k)\}$  for every  $k$ -ary relation  $R$  in  $\Sigma_2$ .

We will use interpretations as a reduction mechanism. The following lemma is crucial for this purpose. It follows by a simple formula rewriting procedure. See, e.g., [20, Thm. 4.3.1].

► **Lemma 10.** *For every  $d$ -dimensional FO-interpretation  $I$  from  $\Sigma_1$  to  $\Sigma_2$ , and FO $[\Sigma_2]$ -formula  $\varphi(\bar{x})$  there exists an FO $[\Sigma_1]$ -formula  $\varphi_I(\bar{x})$  such that for every  $\Sigma_1$ -structure  $G$  and tuple  $\bar{a} \in V(I(G))^{|\bar{x}|}$  we have  $I(G) \models \varphi(\bar{a})$  if and only if  $G \models \varphi_I(\bar{a})$ .*

The above lemma also holds for MSO- and CMSO-interpretations, when restricted to a single dimension. See, e.g., [6, Thm. 7.10]. Higher dimensions are problematic because MSO and CMSO cannot quantify over sets of tuples.

► **Lemma 11.** *Fix  $\mathcal{L} \in \{\text{MSO}, \text{CMSO}\}$ . For every 1-dimensional  $\mathcal{L}$ -interpretation  $I$  from  $\Sigma_1$  to  $\Sigma_2$ , and  $\mathcal{L}[\Sigma_2]$ -formula  $\varphi(\bar{x})$  there exists an  $\mathcal{L}[\Sigma_1]$ -formula  $\varphi_I(\bar{x})$  such that for every  $\Sigma_1$ -structure  $G$  and tuple  $\bar{a} \in V(I(G))^{|\bar{x}|}$  we have  $I(G) \models \varphi(\bar{a})$  if and only if  $G \models \varphi_I(\bar{a})$ .*

For an interpretation  $I$  and a class  $\mathcal{C}$ , we write  $I(\mathcal{C})$  for the class  $\{I(G) : G \in \mathcal{C}\}$ . We say  $\mathcal{C}$   $d$ -dimensionally  $\mathcal{L}$ -interprets a class  $\mathcal{D}$ , if there exists a  $d$ -dimensional  $\mathcal{L}$ -interpretation  $I$  such that  $\mathcal{D} \subseteq I(\mathcal{C})$ . We have already seen in the introduction, that the class of all paths is MSO-unstable. Lemma 11 lifts instability to all classes that 1-dimensionally MSO-interpret it.

► **Lemma 12.** *Fix  $\mathcal{L} \in \{\text{MSO}, \text{CMSO}\}$ . Every class that 1-dimensionally  $\mathcal{L}$ -interprets the class of all paths is  $\mathcal{L}$ -unstable.*

**Flips.** Fix a graph  $G$  and a partition  $\mathcal{P}$  of its vertices. For every vertex  $v \in V(G)$  we denote by  $\mathcal{P}(v)$  the unique part  $Q \in \mathcal{P}$  satisfying  $v \in Q$ . Let  $F \subseteq \mathcal{P}^2$  be a symmetric relation. We define  $H := G \oplus (\mathcal{P}, F)$  to be the graph with vertex set  $V(G)$ , and edges defined by the following condition, for distinct  $u, v \in V(G)$ :

$$uv \in E(H) \Leftrightarrow \begin{cases} uv \notin E(G) & \text{if } (\mathcal{P}(u), \mathcal{P}(v)) \in F, \\ uv \in E(G) & \text{otherwise.} \end{cases}$$

We call  $H$  a  $\mathcal{P}$ -flip of  $G$ . If  $\mathcal{P}$  has at most  $k$  parts, we say that  $H$  a  $k$ -flip of  $G$ . Flips are reversible:  $(G \oplus (\mathcal{P}, F)) \oplus (\mathcal{P}, F) = G$ ; and hereditary: for every  $k$ -flip  $H$  of  $G$  and  $A \subseteq V(G)$ ,  $H[A]$  is also a  $k$ -flip of  $G[A]$ .

It will be convenient to work with flip-partitions that are minimal in the following sense.

► **Definition 13.** *Let  $H$  and  $G$  be two graphs on the same vertex set. We call a partition  $\mathcal{P}$  of  $V(G)$  and a relation  $F \subseteq \mathcal{P}^2$  irreducible  $(H, G)$ -flip-witnesses if*

- $H = G \oplus (\mathcal{P}, F)$ ,
- $H$  is not a  $(|\mathcal{P}| - 1)$ -flip of  $G$ ,
- $F$  does not include  $(Q, Q)$  for any part  $Q \in \mathcal{P}$  with  $|Q| = 1$ .

Clearly for every two graphs  $H$  and  $G$  on the same vertex set, there exist irreducible  $(H, G)$ -flip-witnesses. In particular, as we work with loopless graphs, every flip-relation  $F$  can be modified to satisfy the last condition stating that no singleton part is flipped with itself. We now argue that the irreducible flip-witnesses are uniquely determined.

► **Lemma 14.** *Let  $H$  and  $G$  be two graphs on the same vertex set and  $\mathcal{P}$  and  $F$  be irreducible  $(H, G)$ -witnesses. For every two distinct parts  $Q_1, Q_2 \in \mathcal{P}$  there exists a part  $Q_\Delta \in \mathcal{P}$  such that  $(Q_1, Q_\Delta) \in F \Leftrightarrow (Q_2, Q_\Delta) \notin F$ . We say  $Q_\Delta$  discerns  $Q_1$  and  $Q_2$ .*

**Proof.** Otherwise, we can merge  $Q_1$  and  $Q_2$ , which contradicts irreducibility. ◀

► **Lemma 15 (♣).** *Let  $H$  and  $G$  be two graphs on the same vertex set, let  $\mathcal{P}$  and  $F$  be irreducible  $(H, G)$ -flip-witnesses, and let  $\mathcal{P}'$  and  $F'$  any partition and relation such that  $H = G \oplus (\mathcal{P}', F')$ . Then  $\mathcal{P}$  is a coarsening of  $\mathcal{P}'$ .*

► **Lemma 16 (♣).** *For every two graphs  $H$  and  $G$  on the same vertex set, the irreducible  $(H, G)$ -flip-witnesses are uniquely determined.*

For every two graph  $H$  and  $G$  on the same vertex set, we call  $\mathcal{P}$  the *irreducible  $(H, G)$ -flip-partition* and  $F$  the *irreducible  $(H, G)$ -flip-relation*, if  $\mathcal{P}$  and  $F$  are the unique irreducible  $(H, G)$ -flip-witnesses, as justified by Lemma 16.

**Shrub-Depth, Distances, and Flatness.** We will not work with the original definition of bounded shrub-depth, as given in [16], and which we omit from this paper. Instead, we will use an equivalent definition called  $\infty$ -flip-flatness [11], which we will define in a moment.

The *length* of a path is the number of edges it contains, i.e.,  $P_t$  has length  $t - 1$ . For a graph  $G$  and two vertices  $u, v \in V(G)$  we define  $\text{dist}_G(u, v)$  to be the length of a shortest path between  $u$  and  $v$  in  $G$ . If no such path exists, then  $u$  and  $v$  are in different connected components of  $G$ , and we define  $\text{dist}_G(u, v) := \infty$ . For  $r \in \mathbb{N}$ , a graph  $G$ , and a vertex  $v \in V(G)$  let  $N_r^G[v] := \{u \in V(G) : \text{dist}_G(u, v) \leq r\}$  be the (*closed*) *radius- $r$  neighborhood* of  $v$ . We drop the superscript  $G$  if it is clear from the context. For  $r \in \mathbb{N}$ , we call a set  $A \subseteq V(G)$  *distance- $r$  independent* in  $G$  if  $\text{dist}_G(u, v) > r$  for every two distinct vertices  $u, v \in A$ . Similarly,  $A$  is *distance- $\infty$  independent* in  $G$  if no two vertices of  $A$  are in the same connected component of  $G$ .

► **Definition 17.** *For  $r \in \mathbb{N} \cup \{\infty\}$  and  $k \in \mathbb{N}$ , a set  $A$  of vertices in a graph is  $(r, k)$ -flip-flat, if there exists a  $k$ -flip  $H$  of  $G$  in which  $A$  is distance- $r$  independent. A graph class  $\mathcal{C}$  is  $r$ -flip-flat, if there are margins  $k_r \in \mathbb{N}$  and  $M_r : \mathbb{N} \rightarrow \mathbb{N}$ , such that for every  $G \in \mathcal{C}$  and  $m \in \mathbb{N}$ , every set  $W \subseteq V(G)$  of size  $|W| \geq M_r(m)$  contains an  $(r, k_r)$ -flip-flat subset of size  $m$ .*

We refer to [24] for an introduction to flatness properties. Crucially, flip-flatness characterizes both monadic FO-stability [10] and bounded shrub-depth [11].

► **Theorem 18** ([10] and [11]). *For every graph class  $\mathcal{C}$ :*

- $\mathcal{C}$  is monadically FO-stable if and only if  $\mathcal{C}$  is  $r$ -flip-flat for every  $r \in \mathbb{N}$ .
- $\mathcal{C}$  has bounded shrub-depth if and only if  $\mathcal{C}$  is  $\infty$ -flip-flat.

### 3 Forbidden Induced Subgraphs

Flipped half-graphs  $H_t$  and flipped  $tP_t$ s were defined in the introduction (Definitions 4 and 5). We say a graph class  $\mathcal{C}$  is *pattern-free*, if there is a  $t \in \mathbb{N}$  such that  $\mathcal{C}$  excludes as induced subgraphs every flipped  $H_t$  and every flipped  $tP_t$ . Our goal is to show the following.

► **Proposition 19.** *Every pattern-free class is  $\infty$ -flip-flat.*

In the full paper, we also give a short proof showing that every class of bounded shrub-depth is pattern-free, using the *SC-depth* characterization of shrub-depth [16].

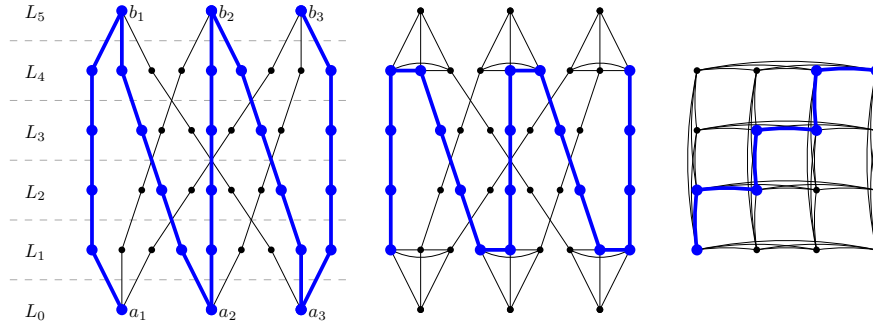
### 3.1 Establishing Monadic FO-Stability

As a first step, we will show that pattern-free classes are monadically FO-stable. This will follow from a known characterization of monadic FO-stability by forbidden induced subgraphs. We first introduce the required definitions.

For  $r \geq 1$ , the *star  $r$ -crossing* of order  $t$  is the  $r$ -subdivision of  $K_{t,t}$  (the biclique of order  $t$ ). The *clique  $r$ -crossing* of order  $t$  is the graph obtained from the star  $r$ -crossing of order  $t$  by turning the neighborhood of each principle vertex (i.e., each original vertex of the  $K_{t,t}$ ) into a clique. Examples are depicted in Figure 5, which also shows how the vertices of each star/clique  $r$ -crossing are partitioned into *layers*  $\mathcal{L} = \{L_0, \dots, L_{r+1}\}$ . A *flipped* star/clique  $r$ -crossing is a graph obtained from a star/clique  $r$ -crossing by performing a flip where the parts of the flip-partition are the layers of the  $r$ -crossing. The following characterization was originally proven in [8], but we use the notation from [24, 11].

► **Theorem 20** ([8]). *A graph class is monadically FO-stable if and only if for every  $r \geq 1$  there exists  $t \in \mathbb{N}$  such that  $\mathcal{C}$  excludes as induced subgraphs*

- all flipped star  $r$ -crossings of order  $t$ , and
- all flipped clique  $r$ -crossings of order  $t$ , and
- all flipped  $H_t$ .



■ **Figure 5** From left to right: the star 4-crossing of order 3, the clique 4-crossing of order 3, and the rook graph of order 4. Highlighted in blue we show how each of the three patterns of order  $k$  contains an induced path on at least  $k$  vertices.

► **Lemma 21.** *For every  $r \geq 1$  and  $t \in \mathbb{N}$ , the star  $r$ -crossing and the clique  $r$ -crossing of order  $t$  both contain an induced  $P_t$ .*

**Proof.** For star  $r$ -crossings with  $r \geq 1$  and for clique  $r$ -crossing with  $r \geq 2$ , this is easy to see in the left/middle panel of Figure 5. For the clique 1-crossing of order  $t$ , note that its vertices  $\{\text{start}(\pi_{i,j}) = \text{end}(\pi_{i,j}) : i, j \in [t]\}$  induce the *rook graph* of order  $t$  (also known as the *cartesian square* of  $K_t$ ). This graph is defined as the graph on vertices  $[t] \times [t]$ , where the vertices  $(i, j)$  and  $(i', j')$  are adjacent if and only if  $i = i'$  or  $j = j'$ . The right panel of Figure 5 shows how  $P_t$  embeds into the rook graph of order  $t$ . ◀

► **Lemma 22** (♣). *For every monadically FO-unstable graph class  $\mathcal{C}$  there exists  $k \in \mathbb{N}$  such that  $\mathcal{C}$  contains as induced subgraphs either*

- a flipped  $H_t$  for every  $t \in \mathbb{N}$ , or
- a  $k$ -flip of  $P_t$  for every  $t \in \mathbb{N}$ .

**Proofsketch.** By Theorem 20 and Lemma 21. See the full paper for details. ◀

► **Lemma 23.** *Every  $k$ -flip of  $(m \cdot k^t)P_t$  contains an induced flipped  $mP_t$ , for every  $k, m, t \in \mathbb{N}$ .*

**Proof.** Choose any  $k$ -flip of  $(m \cdot k^t)P_t$  and let  $\mathcal{P}$  be a witnessing partition of size at most  $k$ . We can color the vertices of the  $P_t$  according to their membership in  $\mathcal{P}$ . This yields at most  $k^t$  different ways to color a  $P_t$ . By the pigeonhole principle, there must be a set  $\mathcal{S}$  of at least  $m$  many  $P_t$  that are colored in the same way. This means for every  $i \in [t]$  all the  $i$ th vertices of paths in  $\mathcal{S}$  are in the same part of  $\mathcal{P}$ . Hence, the paths in  $\mathcal{S}$  induce a flipped  $mP_t$ . ◀

By cutting a long path into multiple shorter ones, we observe that  $P_{t'}$  contains an induced  $mP_t$  for  $t' := m \cdot (t + 1)$ . Combining this observation with Lemma 23 yields the following Corollary 24. The combination of Corollary 24 and Lemma 22 yields Proposition 25.

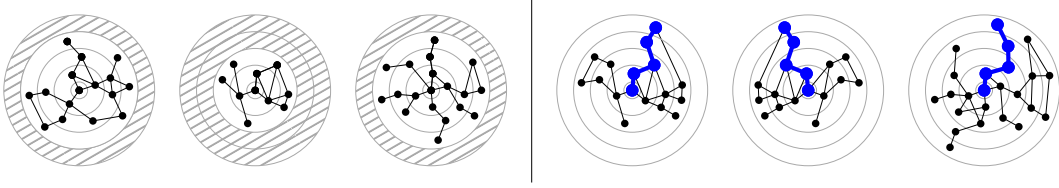
► **Corollary 24.** *Every  $k$ -flip of  $P_{t'}$  contains an induced flipped  $tP_t$  for every  $k, t \in \mathbb{N}$  and  $t' := t \cdot k^t \cdot (t + 1)$ .*

► **Proposition 25.** *Every pattern-free class is monadically FO-stable.*

### 3.2 Proving $\infty$ -Flip-Flatness

► **Lemma 26.** *For every  $m, t \in \mathbb{N}$ , graph  $G$ , and distance- $2t$  independent set  $A$  of size  $2m$  in  $G$  either*

- *a size  $m$  subset of  $A$  is distance- $\infty$  independent in  $G$ , or*
- *$G$  contains an induced  $mP_t$ .*



■ **Figure 6** The two cases from Lemma 26. The circles are the layering of a BFS tree around the vertices of  $A$ . There is a large subset  $A' \subseteq A$  whose outermost layers are either all empty (left panel), or all non-empty (right panel). Therefore, the vertices of  $A'$  are either in pairwise different connected components, or the vertices of  $A'$  are the endpoints of pairwise vertex-disjoint long induced paths.

**Proof.** By the assumption of the lemma, the radius- $(t - 1)$  neighborhoods of the vertices in  $A$  are pairwise disjoint, and every edge is incident to at most one of these neighborhoods. For each  $a \in A$ , let  $X(a) \subseteq N_{t-1}[a]$  be the set of vertices that are at distance exactly  $t - 1$  from  $a$  in  $G$ . By the pigeonhole principle, there is a set  $A' \subseteq A$  of size  $m$  such that either  $X(a)$  is empty for all  $a \in A'$  or  $X(a)$  is non-empty for all  $a \in A'$  (see Figure 6). In the first case this means for each  $a \in A'$ ,  $N_{t-1}[a]$  contains the entire connected component of  $a$  in  $G$ , and  $A'$  is distance- $\infty$  independent. In the second case for each  $a \in A'$ ,  $N_{t-1}[a]$  contains an induced  $P_t$  (which starts at  $a$  and ends in  $X(a)$ ), and  $G$  contains an induced  $mP_t$ . ◀

Using Lemma 23, the previous lemma lifts to flipped graphs.

► **Corollary 27.** *For every  $k, m, t \in \mathbb{N}$ , graph  $G$ ,  $k$ -flip  $H$  of  $G$ , and distance- $2t$  independent set  $A$  of size  $2k^t \cdot m$  in  $H$  either*

- *a size  $k^t \cdot m$  subset of  $A$  is distance- $\infty$  independent in  $H$ , or*
- *$G$  contains an induced flipped  $mP_t$ .*

We are now ready to prove Proposition 19, which we restate for convenience.

► **Proposition 19.** *Every pattern-free class is  $\infty$ -flip-flat.*

**Proof.** Fix a pattern-free class  $\mathcal{C}$ . Then there is  $t \in \mathbb{N}$  such that  $\mathcal{C}$  contains no induced flipped  $tP_t$ . By Proposition 25,  $\mathcal{C}$  is monadically FO-stable, and by Theorem 18, also  $r$ -flip-flat for some margins  $k_r$  and  $M_r$  for every  $r \in \mathbb{N}$ . We claim that  $\mathcal{C}$  is  $\infty$ -flip-flat with margins  $k_\infty := k_{2t}$  and  $M_\infty(m) := M_{2t}(\max(2m, 2k_{2t}^t \cdot t))$ . Choose any  $G \in \mathcal{C}$ ,  $m \in \mathbb{N}$ , and  $W \subseteq V(G)$  of size  $|W| \geq M_\infty(m)$ . By flip-flatness, there is a  $k_\infty$ -flip  $H$  of  $G$  in which a size  $\max(2m, 2k_{2t}^t \cdot t)$  subset  $A \subseteq W$  is distance- $2t$  independent. By Corollary 27, either

- a size  $\max(2m, 2k_{2t}^t \cdot t)/2 \geq m$  subset of  $A \subseteq W$  is distance- $\infty$  independent in  $H$ , or
- $G$  contains an induced flipped  $m'P_t$  for some  $m' \geq \max(2m, 2k_{2t}^t \cdot t)/(2k_{2t}^t) \geq t$ .

The second case contradicts pattern-freeness of  $\mathcal{C}$ , so we have successfully established  $\infty$ -flip-flatness for  $\mathcal{C}$ . ◀

## 4 Interpreting Paths

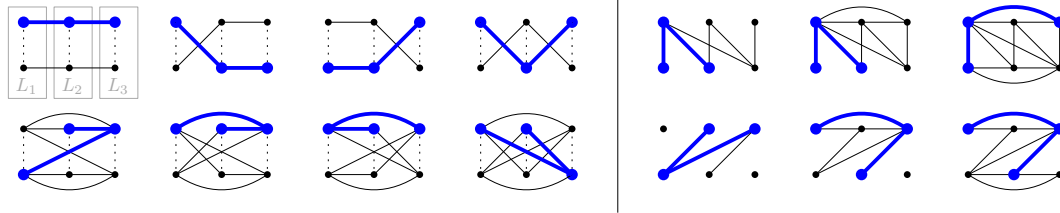
In this section we will prove the following.

► **Proposition 28.** *There is a 1-dimensional FO-interpretation  $I$  with the following property. If  $\mathcal{C}$  is a hereditary graph class that contains for every  $t \in \mathbb{N}$  either a flipped  $H_t$  or a flipped  $tP_t$ , then the class of all paths is contained in  $I(\mathcal{C})$ .*

Notably, a single interpretation works for all classes that contain these patterns. We first show that the paths of length at most 3 appear already as induced subgraphs.

► **Lemma 29.** *Every flipped  $2P_3$  and every flipped  $H_3$  contains  $P_3$  as an induced subgraph.*

**Proof.** Exhaustive case distinctions are depicted in Figure 7. ◀

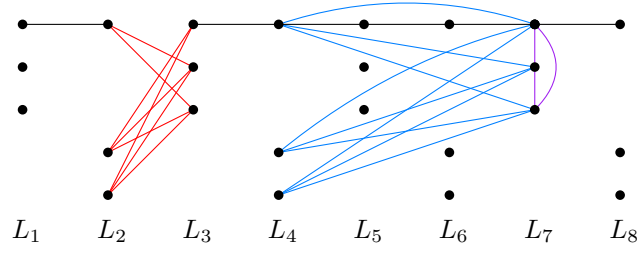


■ **Figure 7** On the left: An enumeration of all flipped  $2P_3$ . To reduce the number of cases, we do not account for flips that flip a layer  $L_i$  with itself. Edges that could possibly be created by such flips are marked as dashed. In each flipped  $2P_3$  we have highlighted an induced  $P_3$  that contains no dashed edges. This proves that every flipped  $2P_3$  contains an induced  $P_3$ . On the right: An enumeration of all flipped  $H_3$ , each with a highlighted induced  $P_3$ .

### 4.1 Interpreting Paths in Flipped $tP_t$ s

Two vertices  $u$  and  $v$  are *twins* in a graph  $G$ , if  $N_1^G[u] \setminus \{u, v\} = N_1^G[v] \setminus \{u, v\}$ . This relation is FO-definable:  $\text{twins}(x, y) := \forall z : (z \neq x \wedge z \neq y) \rightarrow (E(z, x) \leftrightarrow E(z, y))$ .

► **Observation 30.** *Let  $H$  be a  $\mathcal{P}$ -flip of  $G$ . Two vertices  $u, v \in V(G)$  that are contained in the same part of  $\mathcal{P}$  are twins in  $G$  if and only if they are twins in  $H$ .*



■ **Figure 8** The induced subgraph of a flipped  $5P_8$  in which the interpretation  $I$  can interpret  $P_8$ . In this example the following layers were flipped:  $L_2$  with  $L_3$  (red),  $L_4$  with  $L_7$  (blue),  $L_7$  with  $L_8$  (purple).

► **Lemma 31.** *There exists an FO-interpretation  $I$  such that for every  $t \geq 4$ , every flipped  $5P_t$  contains an induced subgraph  $H$  such that  $I(H) = P_t$ .*

**Proof of Lemma 31.** By assumption  $\mathcal{C}$  contains a graph  $H'$  that is an  $\mathcal{L}$ -flip of  $5P_t$ , where  $\mathcal{L}$  is the usual partition of the vertices of  $5P_t$  into layers. We will interpret  $P_t$  in the induced subgraph  $H := H'[A]$  where  $A := X \cup S$  with

$$X = \{(1, i) : i \in [t]\} \quad \text{and} \quad S := S_1 \cup \dots \cup S_t \quad \text{and} \quad S_i := \begin{cases} \{(2, i), (3, i)\} & \text{if } i \text{ is odd,} \\ \{(4, i), (5, i)\} & \text{if } i \text{ is even.} \end{cases}$$

See Figure 8 for an example. In the non-flipped setting, taking the subgraph of  $5P_t$  induced on the same vertex set  $A$  yields a graph  $G := (5P_t)[A]$  that is the disjoint union of a single  $P_t$  (formed by the  $X$  vertices) and an independent set of size  $2t$  (formed by the  $S$  vertices). Observe that  $H$  is an  $\mathcal{L}|_A$ -flip of  $G$ , where  $\mathcal{L}|_A$  is the restriction of  $\mathcal{L}$  to  $A$ . Let  $\mathcal{Q}$  and  $F \subseteq \mathcal{Q}^2$  be the irreducible  $(H, G)$ -flip-witnesses such that  $H = G \oplus (\mathcal{Q}, F)$ . By Lemma 15,  $\mathcal{Q}$  is a coarsening of  $\mathcal{L}|_A$  and, by Lemma 14, for every two distinct parts  $Q_1, Q_2 \in \mathcal{Q}$  there exists a discerning part  $Q_\Delta \in \mathcal{Q}$  such that  $(Q_1, Q_\Delta) \in F \Leftrightarrow (Q_2, Q_\Delta) \notin F$ . In order to interpret  $P_t$  from  $H$ , it suffices to establish the following two claims.

► **Claim 32.** The set  $X$  is definable in  $H$ . This means there is a formula  $\delta(x)$  such that for all  $v \in V(H) = A$  we have  $H \models \delta(v) \Leftrightarrow v \in X$ .

► **Claim 33.** The edges of  $G[X]$  are definable in  $H$ . This means there is a formula  $\varphi(x, y)$  such that for all  $u, v \in X$  we have  $H \models \varphi(u, v) \Leftrightarrow G \models E(u, v)$ .

If we define the interpretation  $I$  using the formulas  $\delta_I := \delta$  and  $\varphi_{I,E} := \varphi$  from Claim 32 and Claim 33, then  $I(H) = P_t$ . It will be clear from the proofs of the two claims, that we can choose  $\delta$  and  $\varphi$  independently of  $t$ , such that  $I$  is the desired interpretation.

**Proof of Claim 32.** We choose  $\delta(x) := \forall y : x \neq y \rightarrow \neg \text{twins}(x, y)$  to be the formula expressing that  $x$  has no twins in  $H$ . We first verify that no vertex in  $S = A \setminus X$  satisfies  $\delta$ . Indeed, each vertex  $u \in S_i \in S$  has a twin  $v$  in  $S_i$  in the graph  $G$ : both vertices have no neighbors at all in  $G$ . As both are in the same part of  $\mathcal{L}|_A$  and its coarsening  $\mathcal{Q}$ , they are also twins in  $H$  by Observation 30.

We next verify that every vertex in  $X$  satisfies  $\delta$ . Fix a vertex  $u \in X$  and let  $Q_1$  be the part of  $\mathcal{Q}$  containing it. Since we assumed  $t \geq 4$ , we know that  $u$  has no twins in  $G$ . Then, again by Observation 30,  $u$  has no twins in  $H$  that are also contained in the same part  $Q_1$ . Now let  $v \in Q_2 \in \mathcal{Q}$  be a vertex contained in a different part. By Lemma 14, there exists a

part  $Q_\Delta$  such that  $(Q_1, Q_\Delta) \in F \Leftrightarrow (Q_2, Q_\Delta) \notin F$ . This means for every vertex  $w \in Q_\Delta$ , its adjacency was flipped towards exactly one of  $u$  and  $v$  from  $G$  to  $H$ . By construction,  $Q_\Delta$  contains at least two vertices from  $S$ , so at least one vertex  $w \in S \setminus \{u, v\} = S \setminus \{v\}$ . As  $w$  is non-adjacent to both  $u$  and  $v$  in  $G$ , we have that  $w$  is adjacent to exactly one of  $u$  and  $v$  in  $H$ , so  $u$  and  $v$  are not twins in  $H$ .  $\triangleleft$

To prove Claim 33 we will recover the partition  $\mathcal{Q}$  using FO. Note that this is not possible for the partition  $\mathcal{L}|_A$ , as it may contain distinct parts which behave the same. This is the reason why we work with the coarsening  $\mathcal{Q}$ , where parts that behave the same have been merged.

Proof of Claim 33. The previous claim also yields a formula  $\sigma(x) := \neg\delta(x)$  that defines the set  $S$  in  $H$ . Consider the formula

$$\pi(x, y) := \forall z : [(z \neq x) \wedge (z \neq y) \wedge \sigma(z)] \rightarrow [E(x, z) \leftrightarrow E(y, z)]$$

expressing that  $x$  and  $y$  have the same neighborhood on  $S \setminus \{x, y\}$ . We claim that for all  $u \in X$  and  $v \in S$ , we have  $H \models \pi(u, v)$  if and only if  $u$  and  $v$  are in the same part of  $\mathcal{Q}$ . If  $u$  and  $v$  are in the same part of  $\mathcal{Q}$ , we can verify that  $H \models \pi(u, v)$  using Observation 30 on the graphs  $H[S \cup \{u, v\}]$  and  $G[S \cup \{u, v\}]$ . If  $u$  and  $v$  are in different parts, we can verify that  $H \not\models \pi(u, v)$  using Lemma 14 as in the proof of the previous claim.

Next, consider the formula

$$\varphi_\oplus(x, y) := \exists z : \sigma(z) \wedge \pi(z, y) \wedge E(x, z)$$

asking whether there exists a vertex  $z \in S$  that is in the same part of  $\mathcal{Q}$  as  $y$  and which is adjacent to  $x$ . We claim that  $\varphi_\oplus$  detects between which vertices of  $X$  the adjacency was flipped from  $G$  to  $H$ . More precisely, for all distinct vertices  $u, v \in X$  we want

$$H \models \varphi_\oplus(x, y) \Leftrightarrow (\mathcal{Q}(u), \mathcal{Q}(v)) \in F.$$

To prove this property, let  $u, v \in X$  be distinct vertices. Assume first  $H \models \varphi_\oplus(u, v)$ . Then there exists a vertex  $z \in S \cap \mathcal{Q}(v)$  which is adjacent to  $u$  in  $H$ . As all the vertices of  $S$  are isolated in  $G$ , this means the adjacency between  $u$  and  $z$  must have been created by the flip, so we have  $(\mathcal{Q}(u), \mathcal{Q}(z) = \mathcal{Q}(v)) \in F$ . Assume now  $(\mathcal{Q}(u), \mathcal{Q}(v)) \in F$ . By construction there exists a vertex  $z \in \mathcal{Q}(v) \cap S$ . As  $z \in S$  is isolated in  $G$ , it must be connected to  $u$  in  $H$ . Then  $z$  witnesses  $H \models \varphi_\oplus(u, v)$ . This proves that  $\varphi_\oplus$  has the desired properties. We can finish the proof of the claim by setting  $\varphi(x, y) := E(x, y) \text{ XOR } \varphi_\oplus(x, y)$ .  $\triangleleft$

Combining the two claims yields the desired interpretation  $I$  satisfying  $I(H) = P_t$ . We again stress that the construction of  $\delta$  and  $\varphi$  was independent of  $t$  (but requires  $t \geq 4$ ). This finishes the proof of the lemma.  $\blacktriangleleft$

## 4.2 Interpreting Paths in Flipped Half-Graphs

► **Lemma 34** ( $\clubsuit$ ). *There exists an FO-interpretation  $I$  such that for every  $t \in \mathbb{N}$ , every flipped  $H_{t+4}$  contains an induced subgraph  $G$  such that  $I(G) = P_t$ .*

**Proofsketch.** Using hereditariness, we can insert twins, which we use to define the two sides  $A$  and  $B$  of the half-graph. We can order the elements of  $A$  by their neighborhoods. The successor relation of the order forms the desired path.  $\blacktriangleleft$

In the full version, we show Proposition 28 by combining Lemmas 29, 31, and 34.

## 5 Monadic Stability

We quickly discuss the two implications from Figure 4 concerning monadic CMSO-stability.

► **Proposition 35** (♣). *Every bounded shrub-depth graph class is monadically CMSO-stable.*

► **Proposition 36** (♣). *Every graph class, that for every  $t \in \mathbb{N}$  contains as induced subgraphs either a flipped  $H_t$  or a flipped  $3P_t$ , is monadically CMSO-unstable.*

Proposition 35 follows mostly from existing techniques. It is known that no class of bounded shrub-depth CMSO-transduces the class of all half-graphs [17]. This implies that no CMSO-formula  $\varphi(x, y)$  on free singleton variables has the order-property on any coloring of a class of bounded shrub-depth. Proving monadic CMSO-stability requires extending this fact to formulas  $\psi(\bar{x}, \bar{y})$  with tuple variables. In the full version, we do so by finitizing a model-theoretic result by Simon [31] (related to [3, 2]) and extending it from FO to CMSO.

To prove Proposition 36, we give an FO-transduction that given a flipped  $3P_t$  produces  $P_t$  (whose vertices are easy to order). This is similar to what we do in Lemma 31, but the more powerful transductions allow us to work in  $3P_t$ s instead of  $5P_t$ s.

## 6 The Expressive Power of MSO

► **Proposition 37** (♣). *MSO is more expressive than FO on every hereditary graph class, that for every  $t \in \mathbb{N}$  contains either a flipped  $H_t$  or a flipped  $tP_t$ .*

**Proofsketch.** Fix a graph class  $\mathcal{C}$  as in the statement. By the pigeonhole principle and hereditariness, it is sufficient to prove the proposition in two separate cases: either  $\mathcal{C}$  contains arbitrarily large flipped  $H_t$  or it contains arbitrarily large flipped  $tP_t$ .

**Case 1.** By Lemma 34, there exists a fixed interpretation  $I$  such that every flipped  $H_{t+4}$  contains an induced subgraph  $H_t^*$  on which  $I$  interprets  $P_t$ . Using this interpretation, we can show that there exists an MSO-sentence  $\varphi_{\text{even}}$  that distinguishing  $H_t^*$  and  $H_{t+1}^*$  for every  $t \in \mathbb{N}$ . The sentence  $\varphi_{\text{even}}$  checks whether the interpreted  $P_t$  has even length, a property that is easily expressible in MSO. To show that there is no FO-sentence equivalent to  $\varphi_{\text{even}}$ , we show that there is a fixed multi-dimensional FO-interpretation  $I'$  that produces  $H_t^*$  from the linear order  $\mathfrak{L}_t = ([t], <)$  of size  $t$ . It is known that for every  $q \in \mathbb{N}$  there is  $t \in \mathbb{N}$  such that no FO-sentence of quantifier rank  $q$  can distinguish  $\mathfrak{L}_t$  and  $\mathfrak{L}_{t+1}$  [23, Thm. 3.6]. The interpretation  $I'$  lifts this result to  $H_t^*$  and  $H_{t+1}^*$ , which proves that no FO-sentence is equivalent to the MSO-sentence  $\varphi_{\text{even}}$  on  $\mathcal{C}$ .

**Case 2.** If  $G$  is a flipped  $tP_t$  with  $t \geq 2$ , we define  $\text{nibble}_1(G)$  to be the graph where the first vertex and last vertex of the first path are deleted, and  $\text{nibble}_2(G)$  to be the graph where the first vertex of the first path and the last vertex of the second path are deleted. Using techniques similar to Lemma 31 we can construct an interpretation  $I$  which reverses the flips on  $G$ . This means  $I(\text{nibble}_1(G))$  is the disjoint union of  $t - 1$  many  $P_t$ s and a single  $P_{t-2}$ ; and  $I(\text{nibble}_2(G))$  is the disjoint union of  $t - 2$  many  $P_t$ s and two  $P_{t-1}$ s. Using this interpretation  $I$ , we can construct an MSO-sentence  $\varphi_{\text{sameParity}}$  which for every graph  $G$ , that is a flipped  $tP_t$ , distinguishes  $\text{nibble}_1(G)$  and  $\text{nibble}_2(G)$ . This sentence checks whether all paths of the interpretation have the same parity: they are either all of even or all of odd length. By construction  $\text{nibble}_1(G) \models \varphi_{\text{sameParity}}$  and  $\text{nibble}_2(G) \not\models \varphi_{\text{sameParity}}$ . To show that no FO-sentence is equivalent to  $\varphi_{\text{sameParity}}$  on  $\mathcal{C}$ , we construct an FO-interpretation  $I'$  that produces  $\text{nibble}_i(G)$  from a coloring of  $\text{nibble}_i(tP_t)$  for both  $i \in \{1, 2\}$ . Here the colors mark the parts of the flip-partition, which means each path is colored the same way. For

every FO-sentence  $\psi$ , there is  $t \in \mathbb{N}$  such that  $\psi$  cannot distinguish between the colorings of  $\text{nibble}_1(tP_t)$  and  $\text{nibble}_2(tP_t)$ : it cannot tell whether two endpoints of the same or of different long paths are missing. Intuitively this is because FO cannot express connectivity. This is easiest to formalize using *Hanf locality* [19, 13], but the classic *Ehrenfeucht-Fraïssé games* also work. Again this inexpressibility can be lifted through  $I'$ , which proves that no FO-sentence is equivalent to the MSO-sentence  $\varphi_{\text{sameParity}}$  on  $\mathcal{C}$ . ◀

## 7 Outlook

In this work, we have characterized graph classes of bounded shrub-depth by forbidden induced subgraphs. This has allowed us to derive further characterizations through logic, for instance by the model theoretic property of MSO-stability, and by comparing the expressive power of FO and MSO. In this last section we will discuss possible future work.

**Stronger Obstructions.** While the obstructions we found were sufficient to achieve our goal of characterizing MSO-stability, we initially had a stronger characterization in mind, which we could neither prove nor refute:

► **Conjecture 38.** *For every graph class  $\mathcal{C}$  of unbounded shrub-depth there exists  $k \in \mathbb{N}$  such that  $\mathcal{C}$  contains as induced subgraphs either*

- *a flipped  $H_t$  for every  $t \in \mathbb{N}$ , or*
- *a  $k$ -flip of  $P_t$  for every  $t \in \mathbb{N}$ .*

As we have seen in Lemma 22, this conjecture holds for every class that is monadically FO-unstable. Even stronger, the  $k$ -flips of  $P_t$ s appearing there can be assumed to be *periodic*. This means, the flip-partition splits the paths in a repeating pattern. This is due to the fact that we have extracted the  $k$ -flips of  $P_t$ , by “snaking” along the known obstructions characterizing monadic FO-stability (see Figure 5). The advantage of these periodic flips is that they can be finitely described: for every monadically FO-unstable class  $\mathcal{C}$  there exists an algorithm that given  $t \in \mathbb{N}$ , returns a size- $t$  obstruction (a flipped  $H_t$  or a  $k$ -flip of  $P_t$ ) that is contained in  $\mathcal{C}$ . We currently do not know whether classes of unbounded shrub-depth admit such algorithms, as we have little control over how the  $tP_t$ s are flipped. Having such a finite description would help to lift algorithmic hardness results from the class of all paths to every hereditary classes of unbounded shrub-depth. One such result is by Lampis, who shows that there is no fpt MSO model checking algorithm for the class of all paths whose runtime dependence is elementary on the size of the input formula, under the complexity theoretic assumption  $E \neq NE$  [22]. Lifting this hardness result to hereditary classes of unbounded shrub-depth would yield another characterization, as it is known that every class of bounded shrub-depth admits elementary fpt MSO model checking [14, 16].

**Relation to the SC-Depth Hierarchy.** Our Theorem 1 only characterizes shrub-depth on the coarse level, distinguishing between classes of bounded and unbounded shrub-depth. One can ask whether the individual levels of the shrub-depth hierarchy (i.e., classes of shrub-depth  $k$  for  $k \in \mathbb{N}$ ) can be characterized similarly. The answer to this is negative, as shrub-depth cannot be meaningfully measured on single graphs, but only on graph classes containing infinitely many arbitrarily large graphs. In particular, every graph class consisting only of a single finite graph has shrub-depth at most 1. A more refined parameter that can also be measured on single graphs is *SC-depth*. On the coarse level, SC-depth and shrub-depth are equivalent: A graph class has bounded shrub-depth if and only if it has

bounded SC-depth [16]. We strongly believe that a more fine-grained analysis of our proofs and the proofs in [28, 11] would yield a functional equivalence between SC-depth and the size of the obstructions we have uncovered in this work. More precisely, for a graph  $G$  define  $\text{obstruction-depth}(G)$  to be the size  $t$  of the largest flipped  $H_t$  or flipped  $tP_t$  contained as an induced subgraph in  $G$ . We conjecture the following.

► **Conjecture 39.** *There are functions  $f, g : \mathbb{N} \rightarrow \mathbb{N}$  such that for every graph  $G$  holds*

$$\text{obstruction-depth}(G) \leq f(\text{SC-depth}(G)) \leq g(\text{obstruction-depth}(G)).$$

**MSO-Dependence.** Our last question concerns the model theoretic notion of *dependence*, a second major dividing line that generalizes stability. Similar to FO-stable classes, also FO-dependent classes have recently been shown to admit nice combinatorial characterizations [11]. This raises the question whether also MSO-dependence can be combinatorially characterized. It is natural to conjecture the following.

► **Conjecture 40.** *For every hereditary graph class  $\mathcal{C}$ , the following are equivalent.*

1.  $\mathcal{C}$  has bounded clique-width (or equivalently bounded rank-width).
2.  $\mathcal{C}$  is MSO-dependent.
3.  $\mathcal{C}$  is monadically MSO-dependent.
4.  $\mathcal{C}$  is CMSO-dependent.
5.  $\mathcal{C}$  is monadically CMSO-dependent.

It is already known that a class has bounded clique-width if and only if it does not CMSO-transduce the class of all graphs [7]. In the full paper we show that this confirms the equivalence (1)  $\Leftrightarrow$  (5) of the conjecture. We remark that both of the two equivalence (1)  $\Leftrightarrow$  (2) and (1)  $\Leftrightarrow$  (3) would imply that every class of unbounded clique-width MSO-transduces the class of all graphs. In turn, this would imply the longstanding Seese’s conjecture stating that the MSO-theory of every graph class of unbounded clique-width is undecidable [30, 6]. This suggests that Conjecture 40 is a tough nut to crack.

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